Why It Matters

Why Misinformation Spreads – Social media algorithms prioritize emotional and engaging content, making false information travel quickly.[1]

Echo Chambers – Platforms show users content similar to their beliefs, reinforcing existing opinions.[1]

Real-World Impact – Misinformation influences elections, spreads health myths, and increases division.[2]

Understanding the Truth Problem Online

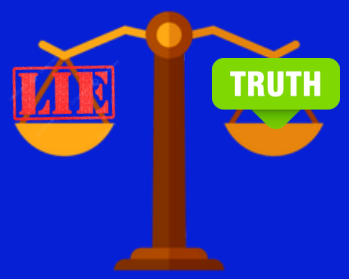
We live in a digital world where information spreads instantly.[2]

But not all information is accurate. Social media platforms use algorithms to decide what users see.[2]

These systems are designed to maximize engagement, not accuracy.[2]

When false content generates strong reactions, it spreads faster than verified facts.[1]

Democracy depends on shared facts.



Grand Challenge: Future of the Internet

Ibrahim Aaryaee
ENG 2003

Engineering Solutions

Algorithmic Auditing

Content Provenance

Credibility Graph Modeling

Platform Transparency Frameworks

What You Can Do?

- Verify before sharing
- Check multiple sources
- Avoid emotionally manipulative headlines
- Ask: Who created this content and why?
- Understand how algorithms shape your feed

Critical thinking is your strongest tool online.

Technology That Can Help

AI Fact-Checking Systems

Automatically detect suspicious claims using machine learning.

Credibility Warning Labels

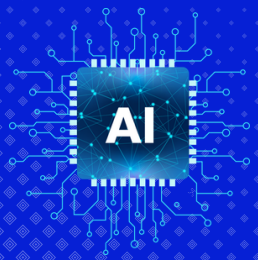
Flag misleading or disputed content.

Algorithm Redesign

Promote diverse viewpoints instead of engagement-only prioritization.

Digital Literacy Education

Teach users how to evaluate sources critically.



References:

[1] M. Cinelli et al., "The echo chamber effect on social media," Proc. Natl. Acad. Sci. U.S.A., vol. 118, no. 9, e2023301118, 2021.

[2] K. L. McLoughlin and W. J. Brady, "Human-algorithm interactions help explain the spread of misinformation," Current Opinion in Psychology, vol. 56, article 101770, Apr. 2024.

[3] U. K. H. Ecker, S. O'Reilly, L. Reid, and J. Chang, "The psychology of misinformation," Psychological Science in the Public Interest, vol. 23, no. 3, pp. 233–316, 2022. (NOT REALLY USED AS MUCH BUT USED REINFORCING IDEAS)

The Systemic Challenge

Digital platforms deploy large-scale recommender systems optimized for engagement metrics such as click-through rate, watch time, and interaction frequency.
These optimization objectives structurally privilege emotionally arousing and identity-reinforcing content.
As a result, misinformation is not merely present — it is algorithmically amplified. [1]

Technical Context

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- Reinforcement learning optimization.
- Graph-based recommendations architectures.
- Attention modeling.
- Large-scale behavioural data collection.

These systems maximize user retention, not epistemic reliability.

Research demonstrates that human cognitive biases interact with algorithmic ranking loops, accelerating misinformation diffusion [2].

Algorithmic Amplification and Information Integrity

ENG 2003
Ibrahim Aaryaee

System-Level Risks

Digital misinformation is not solely a content problem- it is a structural systems problem.

- Epistemic fragmentation across digital communities
- Reinforced ideological clustering through homophily
- Asymmetric misinformation cascades
- Electoral and democratic vunerability.
- Opaque platform governance structures

Engagement-optimized archetectures externalize epstemic costs while internaling attention-based revenue.

Engineering Interventions

Algorithmic Auditing

Independant evaluation of ranking systems for amplification bias and content distortion.

Objective Function Redesign

Shift optimization goals from engagement-only metrics toward diversity-aware and trust-weighted ranking signals.

Credibility Graph Integration

Incorporate source reliability scoring directly into recommendation pipelines.

Content Provenance Infrastructure

Use cryptopgrahic watermarking and metadata verification to trace media origin.

References:

[1] M. Cinelli et al., "The echo chamber effect on social media," Proc. Natl. Acad. Sci. U.S.A., vol. 118, no. 9, e2023301118, 2021.

[2] K. L. McLoughlin and W. J. Brady, "Human-algorithm interactions help explain the spread of misinformation," Current Opinion in Psychology, vol. 56, article 101770, Apr. 2024.

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Artificial Intelligence Disclosure.

For this assignment I used chatgpt to help shorten the data that I had collected from my three sources because it simply was too long to fit entirely on the space provided on the brochure templates that I was using and it became really awkward to fit in the images that I wanted to convey my message. Nothing was AI generated on my thing, it was just shorting my original writing and narrowing it down to a point where I could fit all the information seamlessly into the page. Everything else was of my own work and nothing else really needed the use of AI. If you have any further questions contact me VIA email.

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